



STRATA

Browse in context.

DOC 05 · APP FLOW

App Flow

Every major journey through Strata — onboarding, the everyday browsing loop, Split View, Freeze, Moments, Rewind, and Private Mode.

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strata — local-first browser
confidential — internal working document

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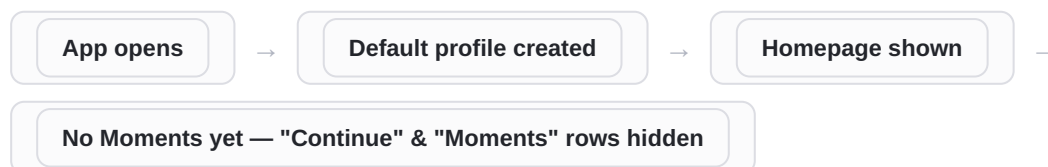
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Flow Map

This document walks every major user journey through Strata end to end — from first launch through everyday browsing, saving and restoring context, and the settings that put the user in control of Continuum. Each flow notes which native module and which Continuum event type it touches, so it can be cross-referenced against the TRD and Backend Schema.

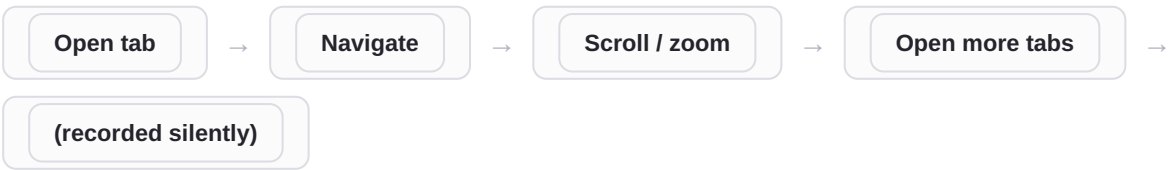


Flow: First Launch



Strata requires no account and no sign-in. On first launch, a default `profiles` row is created locally, `~/.strata/strata.db` is initialized (see Backend Schema §1), and the user lands on the homepage. Because there is no browsing history yet, the Continue and Moments rows on the homepage are omitted rather than shown empty — Strata never shows an empty state that implies a missing feature.

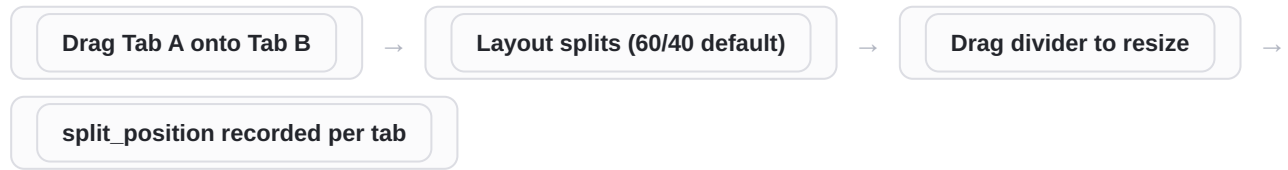
Flow: Everyday Browsing



This is the flow that runs underneath everything else. Every tab creation, navigation, scroll (debounced), and zoom change is written by **EventRecorder** as a discrete event, and **StateCollector** periodically checkpoints the current `PageState` per tab. None of this is visible to the user — no progress indicator, no "saving..." toast. The product's promise is that this happens for free, in the background, with negligible UI impact (TRD §6).

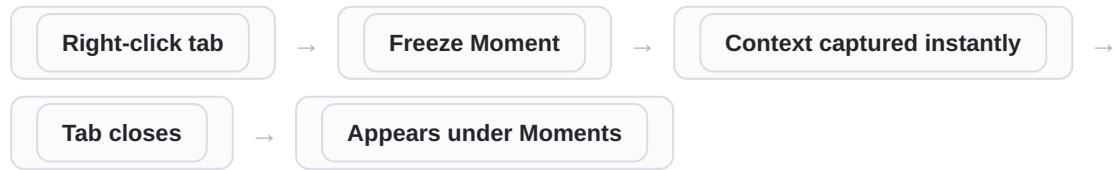
User action		Continuum (silent)
Open tab	→	TAB_CREATED event
Type URL, Enter	→	NAVIGATION event
Scroll	→	debounce 500-1000ms → state checkpoint
Zoom in/out	→	ZOOM_CHANGED event
Drag tab onto tab	→	SPLIT_CREATED event

Flow: Creating a Split View



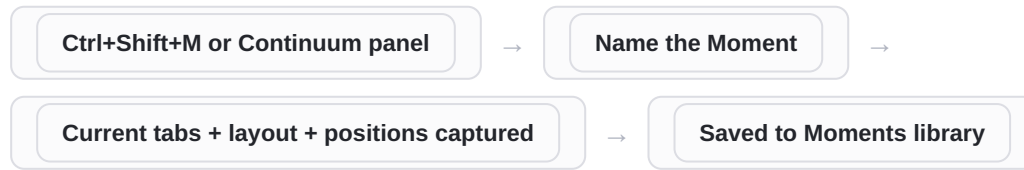
The resulting split ratio is stored per tab in `page_states.split_position` (Backend Schema §3) so that it becomes part of any Moment saved afterward — restoring later reconstructs the exact same pane widths, not just "two tabs, side by side."

Flow: Freezing a Tab



Freeze is the "I don't need this open right now, but I'm not done with it" action. It differs from an explicit Save Moment only in framing — a Freeze auto-names itself from the tab's title/domain and can be renamed later; a Save Moment always prompts for a name up front (see §6).

Flow: Saving a Named Moment



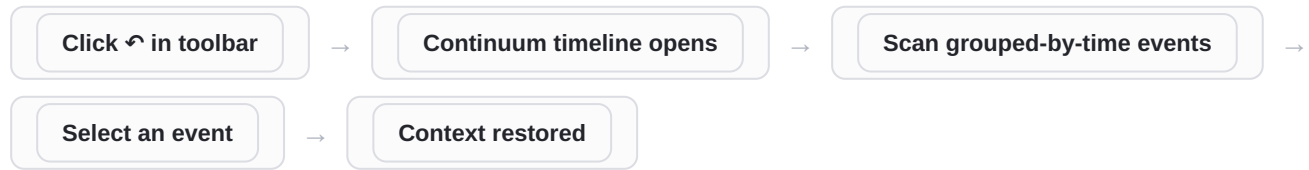
Example — *React Authentication Research*:

```
Moment: React Authentication Research
Created: September 2, 2026 · 8:42 PM
```

Tabs:	Layout:
— React Documentation	— React Docs 60%
— Auth0 Documentation	— GitHub 40%
— GitHub Repository	
— Stack Overflow	Page positions:
— YouTube	— React Docs 72%
	— Auth0 43%
	— GitHub 91%

Unlike Freeze, tabs are *not* automatically closed after Save Moment — the user keeps working and can close them manually whenever they're ready, or leave them open indefinitely with the Moment as a durable checkpoint.

Flow: Rewind & Timeline Navigation



CONTINUUM

TODAY

21:42	GitHub	browser-project
21:36	React Docs	Authentication
21:19	Stack Overflow	OAuth question
20:54	YouTube	React tutorial
20:31	Google	React authentication

Rewind reads from `navigation_events` (append-only) rather than from Moments — it's the raw, chronological record, always available even for browsing that was never explicitly saved. Selecting an entry restores that browsing context as closely as possible, subject to the browser-owned vs. page-owned distinction (PRD §6).

Flow: Restoring a Moment



Restoring Moment...

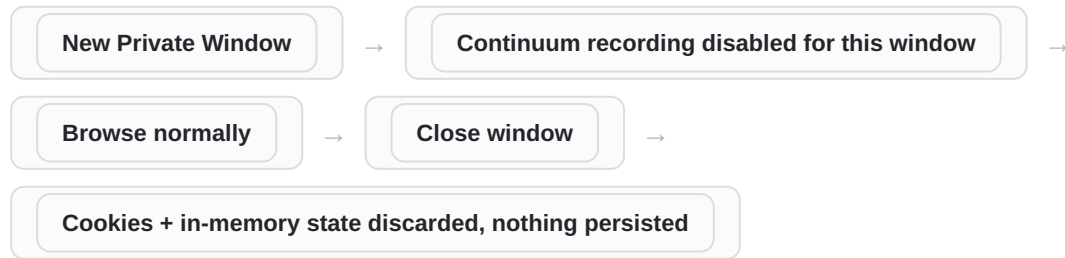
- [GitHub](#)
- [Documentation](#)
- [YouTube](#)
- [Stack Overflow](#)

RestoreManager reads the Moment's `moment_tabs` and associated `page_states`, recreates each tab via BrowserManager/TabManager in original order, reapplies the recorded `split_position` per pane, navigates each tab to its recorded URL, and — once each page has loaded — applies scroll and zoom. Browser-owned fields (URL, scroll, zoom, layout) restore reliably; page-owned fields (form input, canvas/app state) restore only where the destination site still exposes them, and are never guaranteed.

FULL DEMO SCENARIO

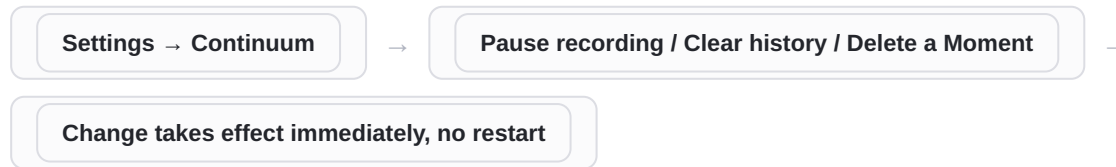
Open Strata → search "How to build a Chromium browser" → open Chromium docs, CEF docs, GitHub, Stack Overflow → split CEF Docs | GitHub → scroll around → Save Moment "Building Strata" → close all tabs → close Strata entirely → reopen → homepage shows "Continue: Building Strata · 4 tabs · 2 hours ago" → click → exact split layout and scroll positions return. This is the flagship demo; see the Implementation Plan for its place in the release sequence.

Flow: Private Browsing



No history, Continuum events, Moments, or cookies persist past the private window's lifetime. Temporary in-memory state may exist while the window is open (for normal tab-to-tab behavior) but is never written to `strata.db`.

Flow: Pausing Continuum & Managing Data



Every control principle from the PRD (§4, "User control") is reachable from one settings screen: pause Continuum entirely, clear history, delete individual Moments, disable state recording, or switch into private browsing. These actions map directly to storage operations described in the Backend Schema (§6, retention & privacy).